MEDIUM · PROBLEM-SOLVING AND DATA ANALYSIS · ANSWER KEY

Problem-Solving and Data Analysis

06

10 answers · Rationale-rich · Teach from doc

ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1

A

A. Plausible values for the average weight of all boxes of oranges filled by the company are between 22.91 pounds and 23.29 pounds.

Choice A is correct. It's given that the estimate for the average weight of all boxes of oranges filled by the company in an 8-hour period is 23.1 pounds, with an associated margin of error of 0.19 pounds. It follows that plausible values for this average weight are between $23.1 - 0.19$ pounds and $23.1 + 0.19$ pounds. Therefore, plausible values for the average weight of all boxes of oranges filled by the company are between 22.91 pounds and 23.29 pounds.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q2

C

C. 300

Choice C is correct. The decimal equivalent of 6% is 0.06. Since increasing the number n by 6% yields the number 318, this situation can be represented by the equation $n(1+0.06)=318$, or $n(1.06)=318$. Dividing both sides of this equation by 1.06 yields $n=300$.

Choice A is incorrect. This is the result when n is increased by 60%, not by 6%. Choice B is incorrect. This is the approximate result of decreasing 318 by 6%. Choice D is incorrect. This is the approximate result of increasing 318 by 6%.

Q3

0.3

Accept also: 3/10

The correct answer is $\frac{3}{10}$. It's given that there are a total of 100 tiles of equal area, which is the total number of possible outcomes. According to the table, there are a total of 30 red tiles. The probability of an event occurring is the ratio of the number of favorable outcomes to the total number of possible outcomes. By definition, the probability of selecting a red tile is given by $\frac{30}{100}$, or $\frac{3}{10}$. Note that 3/10 and .3 are examples of ways to enter a correct answer.

Q4

C

C. 0.22

Choice C is correct. A line in the *xy*-plane that passes through the points $\left(x_1, y_1\right)$ and $\left(x_2, y_2\right)$ has a slope of $\frac{y_2 - y_1}{x_2 - x_1}$. The line of best fit shown passes through the points with approximate coordinates $(0, 2.5)$ and $(10, 4.7)$. It follows that the slope of the line of best fit is approximately $\frac{4.7 - 2.5}{10 - 0}$, which is equivalent to $\frac{2.2}{10}$, or 0.22. Therefore, of the given choices, 0.22 is closest to the slope of the line of best fit.

Choice A is incorrect. This is closest to the negative y-coordinate of the y-intercept, not to the slope, of the line of best fit.

Choice B is incorrect. This is closest to the negative slope, not to the slope, of the line of best fit.

Choice D is incorrect. This is closest to the y-coordinate of the y-intercept, not to the slope, of the line of best fit.

Q5

A

A. mode < median < mean

Choice A is correct. The mode is the data value with the highest frequency. So for the data shown, the mode is 18. The median is the middle data value when the data values are sorted from least to greatest. Since there are 20 ages ordered, the median is the average of the two middle values, the 10th and 11th, which for these data are both 19. Therefore, the median is 19. The mean is the sum of the data values divided by the number of the data values. So for these data, the mean is

$$\frac{(18\times6)+(19\times5)+(20\times4)+(21\times2)+(22\times1)+(23\times1)+(30\times1)}{20}=20.$$

Since the mode is 18, the median is 19, and the mean is 20, $\text{mode} < \text{median} < \text{mean}$.

Choices B and D are incorrect because the mean is greater than the median. Choice C is incorrect because the median is greater than the mode.

Alternate approach: After determining the mode, 18, and the median, 19, it remains to determine whether the mean is less than 19 or more than 19. Because the mean is a balancing point, there is as much deviation below the mean as above the mean. It is possible to compare the data to 19 to determine the balance of deviation above and below the mean. There is a total deviation of only 6 below 19 (the 6 values of 18); however, the data value 30 alone deviates by 11 above 19. Thus the mean must be greater than 19.

Q6

A

A. 7

Choice A is correct. It's given that the object has a mass of 168 grams and a volume of 24 cubic centimeters. Dividing the mass, in grams, of the object by the volume, in cubic centimeters, of the object gives the density, in grams per cubic centimeter, of the object. It follows that the density of the object is $\frac{168 \text{ grams}}{24 \text{ cubic centimeters}}$, which is equivalent to $\frac{168}{24}$ grams per cubic centimeter, or 7 grams per cubic centimeter.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q7

A

A. It is plausible that the actual mean value of the variable for the population is between 19.5 and 21.5.

Choice A is correct. It's given that based on a random sample from a population, the estimated mean value for a certain variable for the population is 20.5, with an associated margin of error of 1. This means that it is plausible that the actual mean value of the variable for the population is between $20.5 - 1$ and $20.5 + 1$. Therefore, the most appropriate conclusion is that it is plausible that the actual mean value of the variable for the population is between 19.5 and 21.5.

Choice B is incorrect. The estimated mean value and associated margin of error describe only plausible values, not all the possible values, for the actual mean value of the variable, so this is not an appropriate conclusion.

Choice C is incorrect. The estimated mean value and associated margin of error describe only plausible values for the actual mean value of the variable, not all the possible values of the variable, so this is not an appropriate conclusion.

Choice D is incorrect. Since 20.5 is the estimated mean value of the variable based on a random sample, the actual mean value of the variable may not be exactly 20.5. Therefore, this is not an appropriate conclusion.

Q8

10

The correct answer is 10. It’s given that the amount of Hanna's food order was $\$50$ and that Hanna gave a tip of 20% of the amount of the bill. 20% of 50 can be calculated as $\frac{20}{100}50$, which yields $\frac{1000}{100}$, or 10. Therefore, the amount, in dollars, of the tip Hanna gave is 10.

Q9

.26

Accept also: `13/50`

The correct answer is $\frac{13}{50}$. It's given that a box contains 13 red pens and 37 blue pens. If one of these pens is selected at random, the probability of selecting a red pen is the number of red pens in the box divided by the number of red and blue pens in the box. The number of red and blue pens in the box is $13 + 37$, or 50. Since there are 13 red pens in the box, it follows that the probability of selecting a red pen is $\frac{13}{50}$. Note that 13/50 and .26 are examples of ways to enter a correct answer.

Q10

C

C. Increasing exponential

Choice C is correct. Because the value of the investment increases each year, the function that best models how the value of the investment changes over time is an increasing function. It’s given that each year, the value of the investment increases by 0.49% of its value the previous year. Since the value of the investment changes by a fixed percentage each year, the function that best models how the value of the investment changes over time is an exponential function. Therefore, the function that best models how the value of the investment changes over time is an increasing exponential function.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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